MILITARY SPECIFICATION

PRE-OILING AND PRESSURE FILL TANK 3-GALLON CAPACITY, TYPE PON-6

This specification has been approved
by the Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 This specification covers one type of 3-gallon capacity, portable, jet oil servicing unit designated Type PON-6.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)
QQ-Z-325	Zinc Plating (Electrodeposited)
PPP-B-591	Boxes, Fiberboard, Wood-Cleated (For Domestic Shipment)
PPP-B-636	Boxes, Fiber
PPP-T-60	Tape; Pressure Sensitive Adhesive, Waterproof; for Packaging and Sealing

Military

MIL-P-78	Plastic Material, Laminated
MIL-P-116	Preservation, Methods of
MIL-B-121	Barrier Material, Greaseproofed, Flexible (Waterproofed)
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification
MIL-P-5515	Pump, Aircraft, Hydraulic, Hand
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-C-7244	Cap and Adapter Unit, Tank Filler
MIL-R-7362	Rubber Sheet, Solid, Molded, and Extruded Shapes, Synthetic Oil Resistant
MIL-L-7808	Lubricating Oil, Aircraft Turbine Engine
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-B-10377	Box, Wood, Cleated, Veneer, Paper Overlaid
MIL-L-19537	Lacquer; Acrylic Nitrocellulose, Gloss

FSC 4930

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STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MS-24475	Nozzle, Pressure Lubricating Oil Servicing
MS-24476	Adapter, Pressure Lubricating Oil Servicing
MS-28720	Filter, Hydraulic Replaceable Element Line Type
MS-29561	Packing, Preformed, O-Ring, Synthetic Lubricant Resistant

PUBLICATIONS

Air Force-Navy Aeronautical Bulletin

143	Specifications and Standards; Use of

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications.- The following documents of the issue in effect on date of invitation for bids form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies of the above publications should be addressed to Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, New York.)

AMERICAN TRUCKING ASSOCIATION
National Motor Freight Classification
Rules and Container Specification

(Application for copies of the above publication should be addressed to The American Trucking Association, 1424 16th Street, N. W., Washington, D. C.)

3. REQUIREMENTS

3.1 Preproduction.- This specification makes provisions for preproduction testing.

3.2 Components.- The unit shall consist of the following major components interconnected as shown in Figure 1:

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DESCRIPTION	SEE REQUIREMENT	
a. Supply tank	3.7	
b. Hand pump	3.8	
c. Flow meter	3.9	
d. Dispensing hose assembly	3.10	
e. Drain bottle assembly	3.11	
f. Filter assembly	3.12	
g. Pressure gage	3.13	
h. Relief valve	3.14	
i. Bleeder valve	3.15	
j. Record card	3.16	Figure 2

3.3 Selection of specifications and standards.- Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with Bulletin 143 except as provided in 3.3.1 and 3.3.2.

3.3.1 Commercial Parts.- Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, such as screws, bolts, nuts, and cotter pins, having suitable properties may be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drawings.

3.3.2 Standard Parts.- With the exception in 3.3.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.4 Material.-

3.4.1 Protective Treatment.- When materials are used in the construction of the unit that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with use, age, or extremes of climatic and environmental conditions shall be avoided. All materials shall be suitable for use with oil conforming to MIL-L-7808.

3.5 Design and Construction.- The unit shall be designed and constructed so that no parts will work loose in service. It shall be built to withstand the stresses, jars, vibrations, and other conditions incident to shipping, storage, installation and service. All parts of the unit shall be new, or the latest design, and of the best quality. The parts shall be of such size material and strength as to properly sustain the maximum load imposed upon them with an adequate factor of safety, maximum working efficiency, and with

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minimum wear during operation. The assembly shall be of clean design with no unnecessary protuberances. The unit also shall have a low center of gravity and shall be capable of being carried and set down in a natural upright position. Provisions shall be made to permit the unit to be firmly grasped and comfortably carried either barehanded or when using heavy gloves. The design shall be such as to allow mass production by low cost methods.

3.5.1 The unit shall be designed to resist upset and shall withstand a force of fifty (50) pounds applied within fifteen (15) degrees of vertical at the center of the pump handle without overturning.

3.5.2 The unit shall be designed to use oil conforming to MIL-L-7808.

3.5.3 Unless otherwise specified, all components requiring O-ring packing shall be provided with O-rings conforming to MS-29561.

3.5.3.1 All gasket material shall conform to Specification MIL-R-7362, or other gasket material impervious to oils noted in paragraph 3.5.2.

3.6 Performance.- At an ambient temperature of plus 70°F, ± 5°F, the pump shall be capable of dispensing oil at a sustained rate of not less than one and one-quarter (1.25) cubic inches per double stroke from the end of a hose located ten (10) feet above the ground. The hose shall be coupled by an adapter conforming to MS-24476-1, to the nozzle of the dispensing hose assembly and shall be an eight (8) foot length size 10 hose. The rate of discharge, when the pump is operated at sixty-eight (68) double strokes per minute, shall be not less than 3 pints per minute. At the start, during, and after a performance demonstration, all lines and hoses shall be full of oil. At the start of a demonstration, all oil in the system shall be at a temperature of plus 70°F, ± 5°F.

3.6.1 Environmental Conditions.- The unit shall not be adversely affected when exposed to the environmental conditions noted below and shall be capable of operation at temperatures from -54°C to ± 40°C.

a. Temperatures ranging from minus 54°C to plus 71°C.
b. Fungus conditions for 28 days.
c. Relative humidity of 95 plus or minus 5 percent at 70° plus or minus 5°C.
d. Atmosphere containing sand and dust particles.
e. Atmosphere containing salt laden moisture.

3.7 Supply Tank.- The supply tank shall be sturdily constructed, have welded or brazed seams, and shall be capable of withstanding an internal hydrostatic pressure of 3 p.s.i. The tank shall have a minimum usable capacity of 3 gallons. The minimum inside clearance diameter on the filler cap opening shall be 3 inches. The tank shall be provided with a drain plug to facilitate cleaning.

3.7.1 Filler cap.- A filler cap, capable of preventing leakage of air or oil shall be provided. The filler cap shall be easily installed or removed by hand or by using a screwdriver on a single screw fastener. Means shall be provided to prevent loss of the cap when opened. The filler cap shall conform to Specification MIL-C-7244, except that the unit shall be suitable for use with oil as per paragraph 3.5.2.

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3.7.2 Air Vent.- A ball-type air vent shall be provided on the tank to prevent pressure or vacuum lock when oil is being pumped into or out of the tank. The vent shall prevent loss of oil if the assembly upsets. The vent shall be provided with means to prevent the entrance of rain, water spray, sand and dust.

3.7.3 Filler Coupling.- An adapter, MS-24476-1, shall be provided to allow filling of the tank by another tank and pump unit which is equipped with the mating nozzle. A plastic cap shall be provided to seal the adapter when not in use. To prevent loss, the cap shall be attached to the nozzle by means of a chain.

3.7.4 Liquid Level Indicator.- The tank shall be provided with a means of determining the approximate level of oil without removal of cap or other closures. A sight gage type consisting of heavy plexiglas may be provided.

3.8 Hand Pump.- A hand-operated, double-acting piston, self-priming type pump shall be incorporated into the design of the unit. The pump shall be capable of delivering a static pressure of 300 psi at minus 65°F with a maximum force of 50 pounds exerted at the center of the grip handle. The pump shall have a displacement of at least one and one-quarter (1.25) cubic inches per double stroke and be in accordance with Class 1500 of MIL-P-5515.

3.9 Flow Meter.- A meter capable of metering oil, when being delivered through the meter at any rate from 0 to 2 g.p.m. shall be provided. The meter shall have two indicators and two graduated circles on the dial. The outer circle of the meter dial shall be graduated in 2-ounce increments, with one revolution of the outer circle indicator representing one quart. The inner circle on the meter dial shall be graduated in quarts and the inner circle indicator shall move one increment for each revolution of the outer circle indicator. The inner circle dial shall be graduated to indicate a total of not less than 14 quarts. The meter dial shall have a diameter of not more than 3-1/2 inches or less than 2-1/2 inches. Means shall be provided to reset the two indicators to their zero flow position. The meter shall have a flow totalizer indicating flow to at least 999 quarts. The meter shall be accurate to within plus or minus 5 percent at a flow of from 1 to 16 pints per minute.

3.10 Dispensing Hose Assembly.- A 6-foot length of size 8 hose suitable for use with oil per paragraph 3.5.2, shall be provided and installed at the discharge end of the flow meter. A nozzle conforming to MS-24475-1 shall be installed at the discharge end of the hose using an intermediate 90-degree elbow. The dispensing hose assembly shall be capable of being coiled and stowed on the top of the unit with the nozzle attached to the adapter on the tank.

3.10.1 Oil Lines.- The oil lines, including piping and connections, shall withstand a test pressure of 600 p.s.i.

3.11 Drain Bottle Assembly.- A one-quart drain bottle assembly shall be provided to catch overflow oil from the aircraft. The assembly shall consist of a one-quart translucent polyethylene bottle graduated in ounce increments attached to a hose with a nozzle, MS-24475-2, on the opposite end. A 90-degree

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fitting shall be inserted between the nozzle and the hose to facilitate use. A plastic cap shall be provided to seal the nozzle opening when the drain bottle assembly is not in use. To prevent loss, the cap shall be attached to the nozzle by means of a chain. Provisions shall be made to stow the drain bottle assembly on the unit. Graduations shall be molded into or etched on the bottle.

3.12 Filter Assembly.- A 10-micron filter shall be installed between the outlet of the pump and the inlet to the flow meter. The filter assembly shall be identical to MS-28720-4 except the case shall not have a bypass and all O-rings and gaskets shall be compatible with oil per paragraph 3.5.2.

3.13 Pressure Gage.- A 0 to 100 p.s.i. pressure gage shall be provided to indicate the pump discharge pressure. The gage shall be graduated in increments of at least 2 p.s.i. and shall have a dial not less than 3 inches in diameter. The gage shall be provided with a clear, crack-resistant, plastic lens at least 1/8-inch in thickness. The gage shall have means of readjustment should recalibration be required. Accuracy of the gage shall be within $\pm$ 2 p.s.i. throughout the range of 0-100 p.s.i.

3.14 Relief Valve.- A relief valve set at 100 $\pm$ 10 p.s.i. shall be provided on the upstream side of the filter to protect the element. Valve discharge shall be returned to the tank.

3.15 Bleeder Valve.- A bleeder valve shall be installed per Figure 1, to relieve the pressure in the system. The valve shall be of the pushbutton type and readily operated by depressing the button.

3.16 Record Card.- A plastic card, 1/16-inch in thickness shall be provided for recording aircraft oil consumption data. The card configuration shall be as shown in Figure 2. The card shall be fabricated from laminated plastic per MIL-P-78, type NDP opaque, with black core and white cover sheets. Markings shall be engraved and shall be essentially as shown. A holder shall be provided on the unit for stowage of the card when not in use.

3.17 Dimensions.- Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the unit, they shall be held or limited accordingly. The general configuration and size of the unit shall be as shown in Figure 3.

3.18 Weight.- The dry weight of the unit less the drain bottle assembly unit shall not exceed 30 pounds.

3.19 Finishes and Protective Coatings.

3.19.1 Aluminum Alloy Parts.- Aluminum alloy parts except parts in contact with the oil shall be covered with a chemical film conforming to MIL-A-8625, except as follows:

3.19.1.1 Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film conforming to MIL-C-5541.

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3.19.2 Plating.- Steel parts requiring plating shall be plated in accordance with one of the following methods:

a. Cadmium plating in accordance with QQ-P-416 of a type and class adequate to achieve the degree of protection required.

b. Zinc plating in accordance with QQ-Z-325.

3.19.3 External Surfaces.- External surfaces requiring paint shall be finished in accordance with MIL-L-19537(Aer). The color shall be orange-yellow color 13538 conforming to Standard FED-STD-595.

3.20 Identification of Product.- Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.21 Workmanship.-

3.21.1 General.- The unit, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, marking of parts and assemblies, welding, brazing, painting, riveting, tightness of screw assemblies, and freedom of parts from burrs and sharp edges.

3.21.2 Riveting.- Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.21.3 Cleaning.- The unit shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips and other foreign material during and after final assembly. Burrs and sharp edges, as well as resin flash that may crumble, shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of Tests.- The inspection and testing of the units shall be classified as follows:

a. Acceptance testing (see 4.2.1)
b. Preproduction testing (see 4.4)

4.2.1 Acceptance Tests.- Acceptance tests shall consist of the following tests performed on each unit:

a. Examination of product (see 4.3.1)
b. Pressure (see 4.3.2)

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4.3 Test Methods.-

4.3.1 Examination of Product.-Each unit shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, marking and finish.

4.3.2 Pressure.-Each tank, including fittings, shall be subjected to an internal hydrostatic pressure of 3 p.s.i. The test shall consist of two applications, each of a duration of not less than two (2) minutes. The tank shall suffer no permanent deformation due to this test. The portion of the system on the discharge side of the pump shall be subjected to an internal hydrostatic pressure of 600 p.s.i. In this test the pressure gage and the relief valve shall be blocked off. Leakage caused by porosity, cracks, faulty welding, or other defects shall be cause for rejection.

4.3.3 Operation.- The unit shall be subjected to flow and operational tests as noted. Each test shall be repeated five (5) times and the entire assembly shall function in a satisfactory manner. The oil used shall be jet lubricating oil conforming to MIL-L-7808.

4.3.3.1 Flow Tests.- The tank shall be filled with exactly three (3) gallons of oil. With the oil and the unit at a temperature of plus 70°F plus or minus 5°F, the dispensing hose of the unit shall be coupled to an 8-foot length of size 10 hose with an MS-24476-1 adapter. With the end of the hose located 10 feet above the ground, the unit shall discharge oil at a rate of at least 3 pints per minute when pumping at a rate of 68 double strokes per minute. At the start, during, and after the test, all lines and hose shall be full of oil. The flow rate shall be based on the time to pump 3 gallons of oil. The meter shall be accurate to within plus or minus 5 percent. Throughout the test, the entire assembly, including pump and meter, shall function in a satisfactory manner.

4.3.3.2 Operational Test.- The tank shall be filled with approximately one gallon of oil at 70°F plus or minus 5°F. With the dispensing hose disconnected and the nozzle located at the level of the base of the unit, the pump shall be actuated 5 times. The system pressure shall not exceed 110 p.s.i. and shall maintain not less than 80 p.s.i. after pumping has ceased. Depression of the relief valve shall bring the system pressure to zero in not more than 4 seconds. The tank shall be drained and an oil supply unit with a rate of flow of 2 g.p.m. shall be connected to the tank adapter with a suitable nozzle. Two and one-half gallons of oil shall be pumped into the tank at a rate of 2 g.p.m. The tank pressure shall not exceed 3 p.s.i. during this test.

4.3.4 Leakage.- The unit, containing two and one-half gallons of oil, shall be turned upside down. No leakage of oil through the air vent or other fittings shall occur.

4.3.5 Environmental Tests.- The unit shall be subjected to the following environmental tests in accordance with the specified procedures of MIL-E-5272.

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Failure of the unit to operate as required during or after each exposure; failure of any component or assembly due to deterioration from exposure; or formation of any undesirable condition which cannot be remedied by routine maintenance procedure shall be cause for rejection.

4.3.5.1 High Temperature.- The high temperature test shall be in accordance with procedure II. There shall be no excessive deterioration as a result of this test. The unit shall be cooled to a temperature of 40°C ± 5°C and at this temperature it shall satisfactorily pass the test specified in 4.3.3.

4.3.5.2 Low Temperature.- The low temperature test shall be in accordance with procedure I. The pump shall deliver a static pressure of 100 p.s.i. with a maximum force of 50 pounds exerted at the center of the handle grip. The flow rate shall be at least 1-1/2 pints per minute when pumping at a rate of 35 double strokes per minute and the meter shall indicate the oil flow within plus or minus 5 percent.

4.3.5.3 Humidity.- The humidity test shall be in accordance with procedure I. There shall be no excessive corrosion resulting from this test.

4.3.5.4 Salt Spray.- The salt spray test shall be in accordance with the procedure specified for a period of 50 hours. There shall be no excessive corrosion resulting from this test.

4.3.5.5 Sand and Dust.- The sand and dust test shall be in accordance with procedure I for a period of 12 hours. No entrance of dust shall be permitted in the oil system.

4.3.5.6 Fungus Test.- The fungus test shall be in accordance with procedure I for a period of 28 days. After exposure, the assembly shall exhibit no effects of fungus attack of a deleterious nature.

4.3.6 Inspection of Preparation for Delivery.- Each unit shall be inspected to determine adequate preparation for delivery in accordance with requirements set forth in Section 5.

4.4 Preproduction Testing.-

4.4.1 Preproduction Test Sample.- The preproduction test sample shall consist of one unit. It shall be tested at a laboratory designated by the procuring activity or at a laboratory approved by the procuring activity upon the request of the contractor or, when so stated in the contract, at the contractor's plant under the supervision of the procuring activity.

4.4.2 Preproduction Tests.- Preproduction tests shall consist of all tests described under 4.3, conducted in the same order.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging.-

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5.1.1 Levels A and B.- The unit shall be preserved and packaged in accordance with MIL-P-116, Method I. The unit shall be wrapped with barrier material conforming to Specification MIL-B-121, Grade A; barrier material to be secured with pressure-sensitive tape conforming to Specification PPP-T-60. Each unit shall be packaged in a fiberboard box conforming to Specification PPP-B-636. The box shall be of a size so that the unit will fit snugly to prevent movement inside the box (See 6.2).

5.1.2 Level C.- The unit shall be preserved and packaged in accordance with the manufacturer's standard practice (See 6.2).

5.2 Packing.-

5.2.1 Level A.- The unit preserved and packaged as specified in paragraph 5.1.1, shall be packed in an exterior shipping container conforming to Specification PPP-B-591, Style A or B; or Specification MIL-B-10377, overseas type (See 6.2).

5.2.2 Level C.- Units shall be packed to insure carrier acceptance and safe delivery to destination in compliance with Uniform Freight Classification Rules or with National Motor Freight Classification Rules and Container Specifications or with other carrier rules and regulations applicable to the mode of transportation (See 6.2).

5.3 Marking.- In addition to any special markings required, packages and shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended Use.- The pre-oiling and pressure fill tank unit covered by this specification is intended for use in servicing jet engines using small quantities of synthetic lubricating oil conforming to Specification MIL-L-7808.

6.2 Ordering Data.- Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Selection of applicable levels of preservation and packaging (See 5.1) and packing (See 5.2).

6.3 Place of Inspection if Different than specified in 4.1.-

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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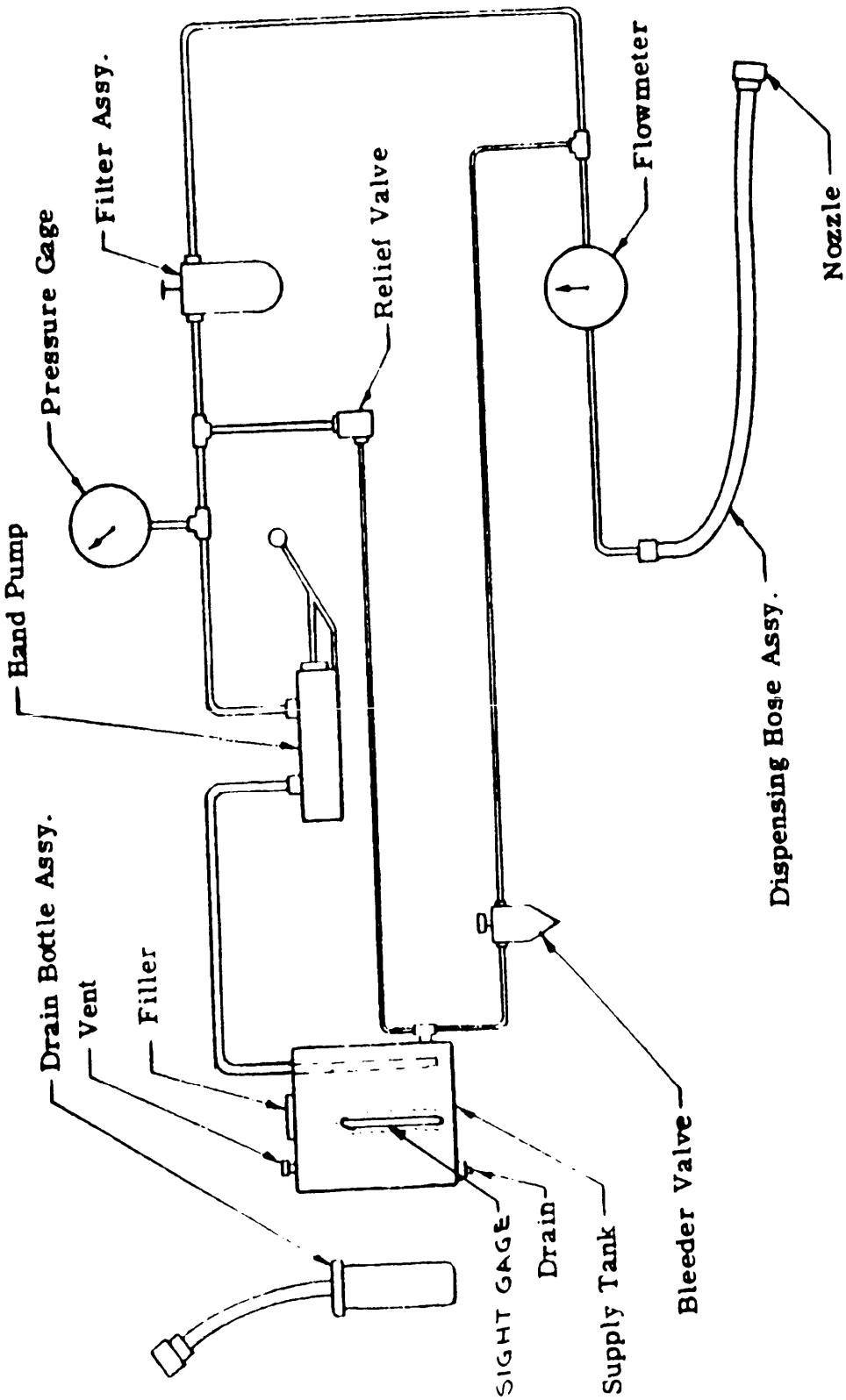


Figure 1. System Schematic

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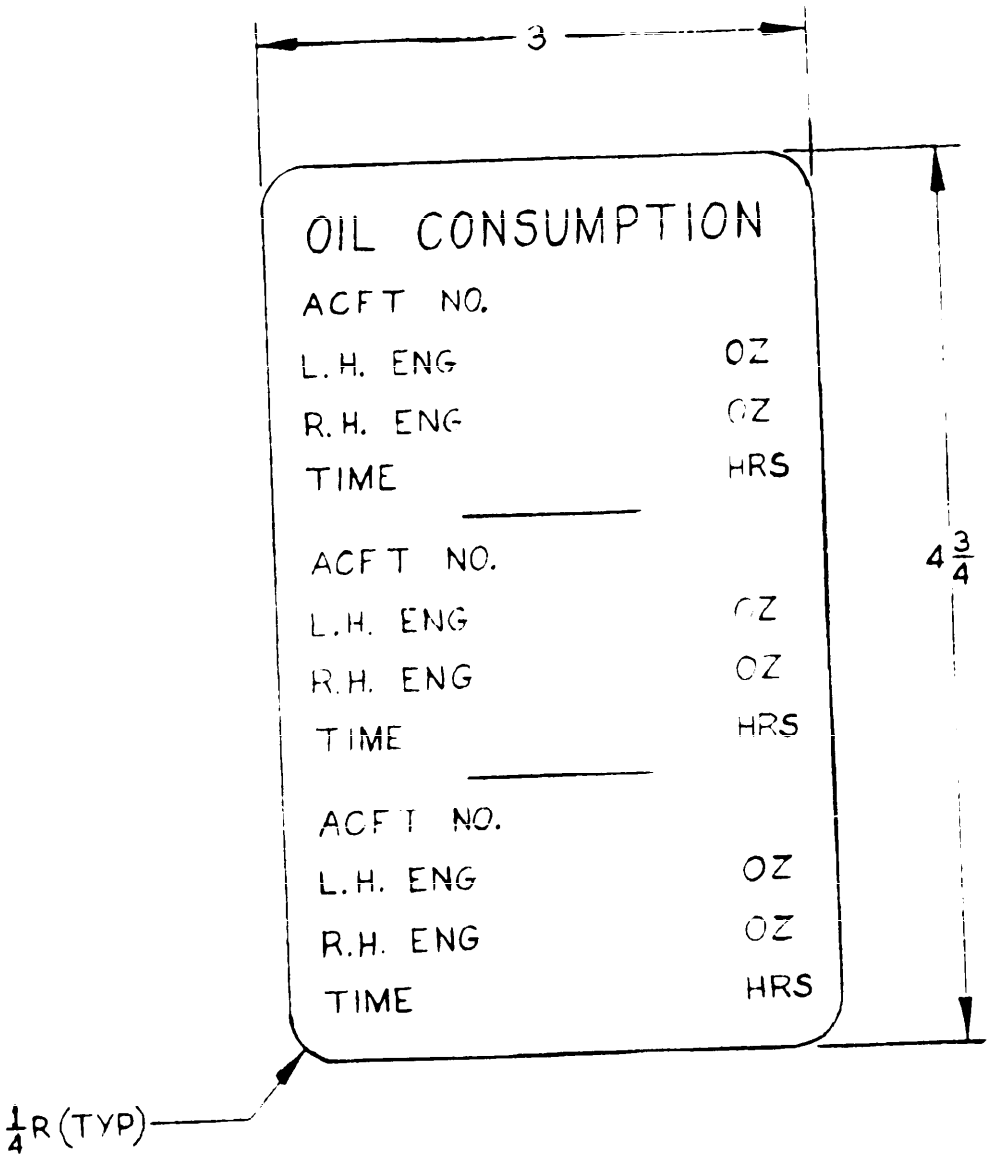


Figure 2. Record Card

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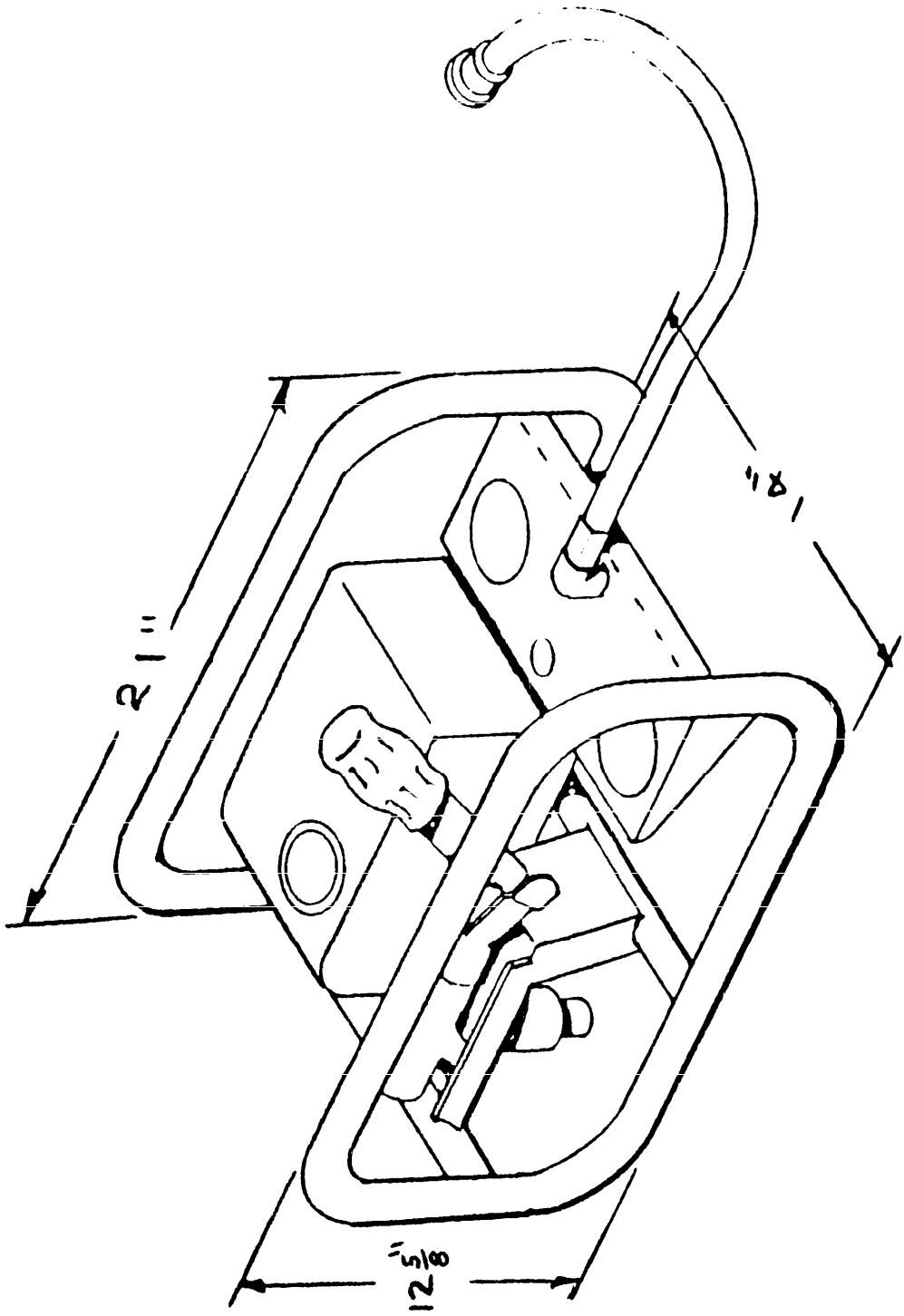


Figure 3. General Configuration and Size

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